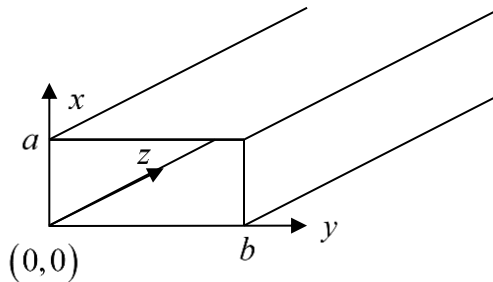
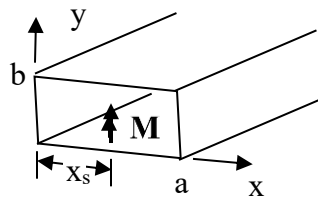
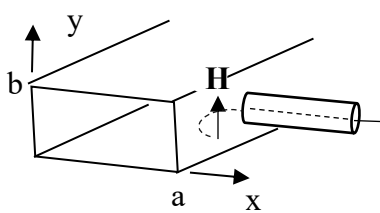


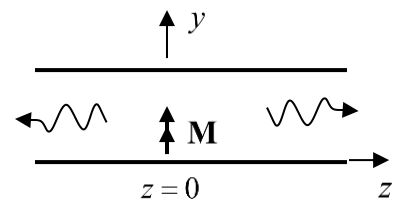
1. (a) Find the TE to x mode electric fields in the rectangular waveguide below.  
 (b) Show the TE to x mode field is a linear combination of the TE and TM to z mode electric fields calculated in the class notes, ie  $\mathbf{E}_{mn}^{TE_x} = A(\mathbf{E}_{mn}^{TE_z} + B\mathbf{E}_{mn}^{TM_z})$ . Find A and B.



2. In order to excite a  $TM_z$  mode in a rectangular waveguide the center conductor of a coax feed line is twisted around and the tip is soldered to the waveguide wall. This essentially creates a vertical magnetic field through the loop or equivalently a magnetic current like the model for a small circular loop antenna. Write down the expressions for the fields in terms of the potentials, state the boundary conditions, hypothesize the potentials in the positive and negative directions based on the boundary conditions, apply boundary conditions at the source, apply orthogonality to find the coefficients in terms of the currents (show the complete orthogonality calculation), write down the final expression for the potential for all modes with the proper summation limits. Give the complete expression for the only  $E_x$  field component. Don't evaluate the integral over the current  $I_m$ .



$$\mathbf{M} = I_m \delta(x - x_s) \delta(z) \hat{\mathbf{y}}$$



side view